



GRIDGUARD AI

Predictive Grid Resilience & Equipment Failure Advisor

Built for the IBM Bob Hackathon • Team Neural Ninjas

26

Monitored Assets

5

Grid Zones

6

Stage ML Pipeline

5

Field Crews

THE PROBLEM

Utilities fly blind between scheduled maintenance checks

Power transformers and substation equipment suffer catastrophic failures that cause expensive outages, severe equipment damage, and risk to public safety.

Utilities still rely primarily on calendar-based maintenance — even though sensor telemetry can reveal degradation long before failure occurs.

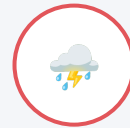
Extreme weather — severe thunderstorms, torrential rain, high wind shear — drastically heightens equipment stress, but asset condition data and meteorological forecasts remain siloed.

TODAY: TWO BLIND SPOTS



Asset Telemetry

Temperature • vibration • partial discharge • oil quality • load — collected, rarely acted on in real time.



Weather Forecasts

Storm and wind data lives in a separate system, disconnected from equipment risk scoring.



Calendar Maintenance

Fixed-interval inspections miss failures that develop between scheduled visits.

THE SOLUTION

GridGuard AI unifies seven data streams into one resilience workflow

01

Asset Health Telemetry

Temperature, vibration, partial discharge, oil quality, load

02

6-Stage ML Failure Inference

Health score → MoG classifier → Isolation Forest → risk

03

Meteorological & Storm Radar

Live NASA POWER satellite weather data

04

Asset Criticality & Topology

26 monitored assets across 5 grid zones

05

Customer & MW Load Exposure

Downstream impact of every asset

06

Field Crew Readiness

Proximity, certifications, equipment — 5 specialized crews

07

IBM Bob AI Advisor

Real-time conversational reasoning across the whole fleet

Prediction → Risk Assessment → Prioritization → Maintenance Interventions → Crew Pre-positioning

ARCHITECTURE

React frontend, FastAPI backend, pluggable ML boundary

React 18 + Vite Frontend — Dark Industrial Operations Center (Tailwind CSS)

▼ REST APIs (/api/*)

FASTAPI BACKEND

- Composite Risk Engine (4-weight formula)
- IBM Bob AI Advisor (fleet-wide reasoning)
- NASA POWER Weather Integration
- Maintenance Dispatch & Crew Optimizer (5 crews)
- Interactive TR-104 Demo Simulator
- 6-Stage ML Pipeline

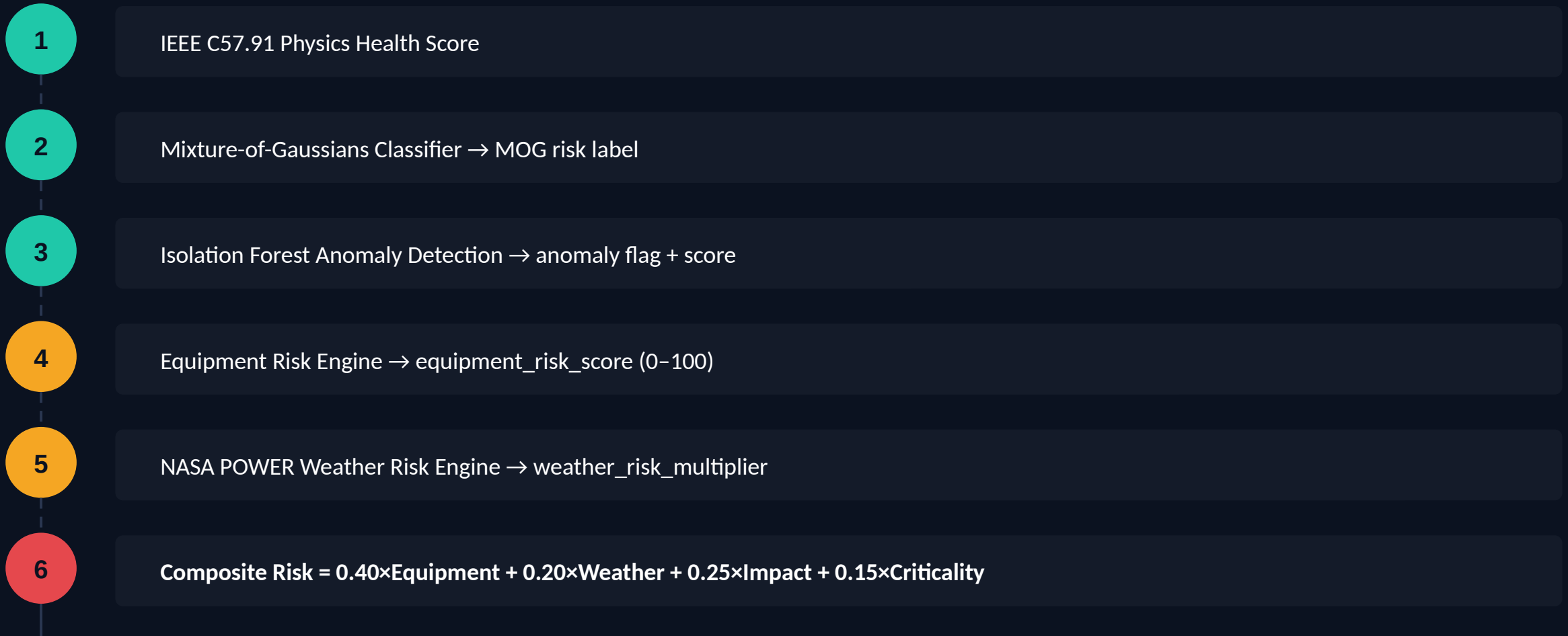
SQLite Database

26 assets • 2,600+ sensor readings • incidents • forecasts • crews

IBM Bob AI API

Granite/fast model — real-time LLM reasoning over the fleet

Six stages turn raw sensor data into a single risk score



A conversational advisor with full fleet context

The AI Operations Advisor (/advisor) is powered by IBM Bob AI using the Granite/fast model. It reasons across all 26 assets and 5 grid zones in real time, and falls back to rule-based heuristics if the API is unavailable.

1

Fleet-wide Q&A

Answers general fleet health, maintenance plan, and crew staging questions with full system context.

2

Asset deep-dives

Provides asset-specific analysis on demand — e.g. “what's the status of TR-104?”

3

Graceful fallback

Falls back to rule-based heuristic reasoning if the API is unavailable.



AI Operations Advisor

“Which assets are highest risk in the Eastern zone right now?”

TR-104 (Naroda) is CRITICAL at 94/100 — partial discharge +31%, storm stress adding +26 pts. Recommend pre-positioning Crew 2, staged 10km out with thermal imaging and oil testing gear.

“What's the fleet-wide maintenance priority this week?”

TR-104 at Naroda Substation — baseline to storm-driven critical alert

STAGE 1

Baseline Nominal Operations

- Normal telemetry
- Risk Score: 42/100
- Health: 72/100 — OPERATIONAL
- Maintenance Queue: Priority #4 (routine)

STAGE 2

Incipient Degradation

- Temp rises +10°C
- Vibration harmonics elevate
- Partial discharge climbs to 28.5 pC
- Risk Score: 68/100 — WARNING
- Priority promoted to #2

STAGE 3

Storm Alert & Critical Spike

- Partial discharge surges +31% (42 pC)
- Winding temp reaches 91.2°C
- Storm: 48.5 mm/h rain, 52 km/h wind
- Composite Risk: 94/100 — CRITICAL
- 82% failure likelihood — Priority #1
- Crew 2 pre-positioned (10km out)

Risk decomposition: Base Equipment Risk (68) + Weather Stress (+26) = Final Risk (94)

Nine views cover the full operations workflow

Dashboard

Grid Health Index, risk distribution, topology map, 24h trend chart

Fleet Inventory

Filter, search, sort, and paginate across all 26 assets

Asset Detail & SCADA

5 live sensor charts, health diagnostics, explainable risk factors

Geospatial Observatory

Full-screen map with Doppler storm radar overlay

Dispatch Planner

Prioritized queue, crew pre-positioning, one-click assignment

Meteorological Intel

5-zone risk table, active alerts, 7-day NASA POWER forecast

Reliability Archive

Historical failure log, root-cause forensics, restoration times

AI Operations Advisor

IBM Bob AI conversational reasoning with evidence-backed guidance

Engine Tuning

Composite risk weight sliders and ML integration status

A production-shaped stack, containerized end to end

FRONTEND

React 18, TypeScript, Vite, Tailwind CSS, Recharts, Leaflet, Lucide React, React Router DOM

BACKEND

Python 3.11, FastAPI, Uvicorn, SQLAlchemy 2.0, Pydantic v2, HTTPX, Pytest

ML / AI

Scikit-learn (Isolation Forest, ExtraTrees), IBM Bob AI API (Granite/fast), IEEE C57.91 physics model

DATABASE

SQLite — 26 pre-seeded deterministic grid assets, 2,600+ sensor readings

WEATHER

NASA POWER satellite API — live meteorological data integration

DEPLOYMENT

Docker, Nginx, Supervisor — single-container production deployment

Tested, documented, and deployable in one command

22/22

Automated backend tests passing
(pytest test_api.py)

1

Command Docker Compose
deployment — docker compose up --
build

3

ML documentation files —
dataset.md, models.md, pipeline.md

0

Frontend changes needed to swap in
a teammate's ML model

PLUGGABLE ML BOUNDARY

The frontend talks exclusively to POST /api/ml/predict. The ML model is isolated behind src/backend/app/services/ml_service.py — swapping in an external model is a one-line .env change (USE_EXTERNAL_ML_SERVICE=true), with zero frontend coupling.

The team behind GridGuard AI



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Prediction before failure. Crews before the storm.

github.com/soni-shashan/bob-ai-hackathon-neural-ninjas

Thank you — Team Neural Ninjas